and liquefaction
111 Hydrogen Hydrogen Water +
Electricity
Fossil/Renewable Electrolysis and
liquefaction
Grid mix electricity LH2_EL_gm_Liquefied
112 Hydrogen Hydrogen Water +

---CHUNK---

SOURCE: MEPC.376(80).pdf
other Electrolysis
and liquefaction
111 Hydrogen Hydrogen Water +
Electricity
Fossil/Renewable Electrolysis and
liquefaction
Grid mix electricity LH2_EL_gm_Liquefied
112 Hydrogen Hydrogen Water +
Electricity
Nuclear Thermochemical
Cycles or
Electrolysis and
liquefaction
Nuclear LH2_EL_n_Liquefied
113 Hydrogen Hydrogen Industrial by-
product hydrogen
 Grid mix electricity LH2_ _ibp_gm _Liquefied
114 Ammonia Ammonia Natural Gas Fossil Methane Pyrolysis
into pure carbon
and hydrogen and
Haber Bosch
process
Grid mix electricity NH3_f_MPO_HB_gm
115 Ammonia Ammonia Natural Gas Fossil Steam Methane
Reformation of
Natural Gas and
Haber Bosch
process
Grid mix electricity NH3_f_SMR_HB_gm
116 Ammonia Ammonia Natural Gas Fossil Steam Methane
Reformation of
Natural Gas with
Point Source
Carbon Capture
(PSCC) and long-
term storage and
Grid mix electricity NH3_f_SMR_HB_CCS_g
m

---CHUNK---

SOURCE: MEPC.376(80).pdf
m
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
Annex 14, page 46

I:\MEPC\80\MEPC 80-17-Add.1.docx
Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
Haber Bosch
process
117 Ammonia Ammonia Coal Fossil Gasification of
Coal and Haber
Bosch process
Grid mix electricity NH3_f_G_HB_gm
118 Ammonia Ammonia Coal Fossil Gasification of
Coal with Carbon
Capture and long-
term storage and
Haber Bosch
process
Grid mix electricity NH3_f_G_HB_CCS_gm
119 Ammonia Ammonia 2nd Gen.
feedstock
Biogenic Gasification Grid mix electricity NH3_b_G_2ndgen_gm_
120 Ammonia Ammonia N2 + H2 N2: separated with
renewable
electricity

H2: produced from
renewable
electricity
Haber Bosch
process
Grid mix electricity NH3_rN2_rH2_HB_gm
121 Ammonia Ammonia N2 + H2 N2: separated with
renewable
electricity

H2: Fossil Steam
Methane
Reformation
Haber Bosch
process

Grid mix electricity NH3_rN2_fH2_HB_gm
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)

---CHUNK---

SOURCE: MEPC.376(80).pdf
121 Ammonia Ammonia N2 + H2 N2: separated with
renewable
electricity

H2: Fossil Steam
Methane
Reformation
Haber Bosch
process
Grid mix electricity NH3_rN2_fH2_HB_gm
RESOLUTION MEPC.376(80) (adopted on 7 July 2023)
GUIDELINES ON LIFE CYCLE GHG INTENSITY OF MARINE FUELS (LCA GUIDELINES) MEPC 80/17/Add.1
Annex 14, page 47

I:\MEPC\80\MEPC 80-17-Add.1.docx
Order Group Fuel type
Feedstock structure Conversion/Production process Fuel Pathway Code
Feedstock
Type
Nature/Carbon
Source Process Type Energy used in the process
122 Ammonia Ammonia N2 + H2 N2: separated with
renewable
electricity

H2: Industrial by-
product hydrogen
Haber Bosch
process
Grid mix electricity NH3_rN2_ibpH2_HB_gm
123 Ammonia Ammonia N2 + H2 N2: separated with
grid mix electricity

H2: Fossil Steam
Methane
Reformation
Thermochemical
Cycles or
Electrolysis
Nuclear NH3_gmN2_fH2_EL_n
124 Ammonia Ammonia N2 + H2 N2: separated with
grid mix electricity

H2: produced from
renewable
electricity
Thermochemical

Cycles or
Electrolysis
Nuclear NH3_gmN2_rH2_EL_n
125 Ammonia Ammonia N2 + H2 N2: separated with
grid mix electricity

H2: Industrial by-
product hydrogen

---CHUNK---

SOURCE: MEPC.376(80).pdf
grid mix electricity

H2: produced from
renewable
electricity
Thermochemical
Cycles or
Electrolysis
Nuclear NH3_gmN2_rH2_EL_n
125 Ammonia Ammonia N2 + H2 N2: separated with
grid mix electricity